

The Depreciation of Human Capital: Evidence from Italy

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Abstract

How does the return to human capital vary with education and gender over the life cycle? Using the Bank of Italy Survey on Household Income and Wealth for 1980–2019 (~100,000 observations), we estimate age–wage and hours–age profiles by ordinary least squares on the conditional expectation function, absorbing cohort and year fixed effects, following Hendricks (2013). Two results emerge. First, a secondary rather than a lower secondary education raises the return to human capital by about 1% per year of age and lowers time spent working by about 0.5%. Second, the gender gap in that return is wider among the higher-educated, at roughly 2.2% per year lower for women. Because women who leave employment are unobserved, selection on the extensive margin biases the female estimates and the gap reported here should be read as a lower bound.¹

¹Total words: 4996

1 Introduction

The Ben-Porath model has been widely used in economic research to explain differences in wage-age profiles resulting from variations in human capital. Human capital encompasses the knowledge, skills, and abilities individuals gain through education and training, enabling them to participate in the labour market and contribute to the economy. While human capital is subject to depreciation over time, individuals can invest in their human capital to improve it. The difference equation that describes human capital includes a depreciation rate, a cohort-specific productivity parameter, a curvature parameter, and age t .

Numerous studies have employed the model of Ben-Porath (1967) to analyse cross-regional differences in wage profiles (Baker and Holsinger, 1997) and the evolution of wage inequality, as in Joshi et al. (2021) and Erosa et al. (2012). However, there is still a need to explore how human capital evolves for individuals with different levels of schooling, with a focus on potential differences in the wage profiles of women. Understanding the evolution of human capital is crucial to identifying potential constraints to long-term economic growth, as shown in Mincer (1981), as a population with higher depreciation rates and poor human capital can limit potential growth. Moreover, identifying different returns on human capital allows us to better understand the incentive to invest in education.

This paper aims to estimate the age-wage profiles and depreciation rates for different education levels while minimising assumptions, following the $\log(y)$ method of Hendricks (2013). Additionally, we examine the impact of gender discrimination on the long-term human capital of women, which can generate different returns to education. The findings of this study will provide valuable insights into the dynamics of human capital accumulation and depreciation in Italy. This research also informs policy interventions aimed at promoting economic growth and development by highlighting potential constraints to long-term growth and the importance of reducing discrimination in the labour market.

Ultimately, this research contributes to the broader literature on the role of education and human capital in economic growth and development, as well as the importance of promoting gender equality in the labour market.

2 Empirical part

The empirical questions that we set out to answer are the following:

1. What is the return in remuneration of human capital for different levels of education?
2. Is there a difference in the remuneration of human capital between men and women?

2.1 The model predictions

The Ben-Porath model suggests that the depreciation rate of human capital is a linear function of time and depends on the level of human capital. This is because individuals with higher human capital face a higher depreciation rate, similar to the relationship between capital and growth in the growth theory. As such, we can expect that individuals with lower education levels, and therefore lower levels of human capital, will experience a lower depreciation rate compared to those with higher levels of education.

In terms of the wage-age profile, individuals with lower education levels will typically invest less in human capital and reach their maximum wage sooner and at a lower level compared to those with higher education levels. This is because individuals with higher levels of education are more likely to acquire skills and knowledge that lead to higher wages and greater career opportunities later in their careers.

2.2 Specification

We use regression on the Conditional Expectation Function (CEF) as a method to estimate the relationship between the dependent variable:

- $\log(wage)_{i,t}$
- $\log(hours)_{i,t}$

(where i indexes the individual and t the age) and the independent variables:

- age, age^2 : the age and the age squared of the individual i of the year y
- $i.cohort$ is the list of dummies of the cohorts

- $i.years$ is the list of dummies for the years class (class size = 5).

while controlling for:

- $educ = \{3, 5\}$: educational level, 3: lower secondary, 5: secondary
- $female = \{0, 1\}$: gender (= 1 if female)

We use four specifications:

1. $\log(wage)_{i,y} = \beta_{age}age_i + \beta_{age^2}age_i^2 + i.cohort + i.years \mid educ = \{3, 5\}$
2. $\log(wage)_{i,y} = \beta_{age}age_i + \beta_{age^2}age_i^2 + i.cohort + i.years \mid educ = \{3, 5\}, female = \{0, 1\}$
3. $\log(\overline{hours})_{i,y} = \beta_{age}age_i + \beta_{age^2}age_i^2 + i.cohort + i.years \mid educ = \{3, 5\}$
4. $\log(hours)_{i,y} = \beta_{age}age_i + \beta_{age^2}age_i^2 + i.cohort + i.years \mid educ = \{3, 5\}, female = \{0, 1\}$

We adopt the specifications given by Hendricks (2013), denoted as (1) and (2). However, we do not include a gender fixed effect in the regression analysis, as we aim to allow gender to affect the estimates of β_{age} and β_{age^2} in a time-varying manner. This approach is similar to that proposed by Joshi et al. (2021). We also refrain from using months of tenure as a proxy for experience, as it may result in collinearity issues. The CEF method has several limitations.

First, it assumes that the age and education variables capture all the relevant dimensions of human capital. Other important factors, such as experience or on-the-job training, are not explicitly considered in the model. Second, the method may suffer from omitted variable bias if there are unobserved characteristics that affect both wages and education levels. Third, the CEF method assumes that the relationship between age, education, and wages is linear, which may not hold in reality.

2.3 Identification strategy

To estimate the depreciation rate of human capital for different levels of education, it is necessary to implement an identification strategy that can isolate the effects of age, education, cohort, and year on wage profiles. This

section describes the identification strategy used in a study to achieve this objective.

Firstly, the assumption is made that cohort effects are constant for all agents in the same 5-year cohort. This is because individuals who are born in the same 5-year period tend to share similar social, economic, and political experiences during their formative years. By assuming that cohort effects are constant across individuals in the same 5-year cohort, it is possible to control for unobserved heterogeneity that is specific to a particular cohort.

Secondly, it is assumed that the initial human endowment is constant within the same cohort. This implies that individuals who are born in the same 5-year cohort have similar levels of human capital at the beginning of their careers, before investing in additional education or training. By assuming that the initial human endowment is constant within the same cohort, it is possible to control for unobserved heterogeneity that is specific to a particular birth cohort.

Similar assumptions apply to year effects, which means that individuals within the same cohort experience the same effects on wages over a period of 5 years. These restrictions on the model are based on some naive assumptions about the behaviour of individuals concerning their education and human capital. For instance, it is reasonable to assume that individuals who are born in the same 5-year cohort would have similar opportunities for education and training. Similarly, individuals who start with the same initial human capital endowment within the same cohort are likely to face similar labour market opportunities over time.

While these assumptions may not hold perfectly in reality, they provide a useful framework for identifying the key factors that drive the age-wage profile and depreciation rates of human capital for different levels of education in Italy. By controlling for cohort and year effects in the analysis, it is possible to isolate the effect of age and education on wages and estimate the rate of human capital depreciation for different groups of individuals.

To estimate the depreciation rate of human capital for different levels of education, a longitudinal panel data approach is used. Specifically, two separate OLS regressions are run for individuals with a lower-secondary education and those with a high-school-level education, respectively. The dependent variable is the natural logarithm of the wage, and the key independent variables are age and age squared. Cohort effects and year effects are controlled by including birth year dummies and calendar year dummies, respectively. Additionally, the wage variable is corrected for inflation using the consumer

price index.

Moreover, to examine gender differences in the evolution of human capital, four separate regressions are performed, conditioned on gender and on the two levels of education, as in (2). To capture the potential variation in gender effects over time, the regression on the conditional expectation function is employed. Additionally, the same $\log(y)$ method is used to estimate the gender gap in the hours-age profile.

Overall, the identification strategy used in this study allows for the separation of the effects of age, education, and cohort on wage profiles and the estimation of the depreciation rate of human capital for different levels of education. By using a longitudinal panel data approach and controlling for individual-specific unobserved heterogeneity and time-varying factors, more accurate estimates of the depreciation rate of human capital can be obtained. This is a crucial variable that affects overall economic growth and development.

3 Data and descriptive statistics

3.1 Data

The data used in this study are obtained from the Survey on Household Income and Wealth of the Bank of Italy, which spans the period from 1980 to 2020 with $\sim 100,000$ observations. The sample comprises employees,² both men and women, born between 1940 and 2020. To expand our sample size, we have divided the population into 15 birth cohorts, where each cohort covers five consecutive birth years. We have also applied the same division for years, where each group includes five years. Table 1 tabulates the distribution of observations across cohorts. The cohort with the highest density of observations is cohort 10, whereas the younger and older cohorts have relatively few observations. The objective of the study is to estimate wage-age profiles, which required the use of employee income reports merged with family and individual statistics. To calculate $\log(wage)_{i,t}$, we employed a series of transformations. Firstly, we summed the monetary and non-monetary components of income. Secondly, we deflated the income figures using the Consumer Price Index (CPI). Thirdly, we divided the income figures by the

²We drop the observations for the self-employed, since including them would raise an issue of under-reporting.

monthly hours worked and multiplied by 13 months to obtain the annual income. Fourthly, we took the logarithm of the annual income figure. Finally, we merged the resulting dataset with individual-level data.

The process of computing $\log(wage)_{i,t}$ was essential for the analysis, as it allowed us to measure the variation in wages over time, which is critical for estimating wage-age profiles. By merging different datasets, we were able to obtain a more comprehensive picture of the factors influencing wage dynamics in Italy.

3.2 Descriptive statistics

The tables presented in this paper (see Tables 2 and 3) provide a detailed overview of the descriptive statistics for two key variables: the logarithm of wages and annual working hours. The tables show the mean, standard error, and t-test for the difference in mean for each variable across four different groups based on education level and gender. Specifically, the groups are defined by individuals with a lower secondary school education and those with a secondary school education, and these subgroups are further differentiated by gender.

The statistical results in Table 2 indicate that individuals with a secondary school education level tend to earn higher wages than those with a lower level of education. Moreover, the wage gap between genders is more pronounced among those with a secondary school education level. In other words, women tend to earn less than men across all education levels, and the difference is wider for those with a higher level of education.

Individuals with lower education tend to work more hours annually than those with higher education, while women tend to work fewer hours than men across all education levels. However, the gender gap in working hours narrows as the level of education increases, according to Goldin and Polachek (1987). It is worth noting that the data analysed covers the period from 1980 to 2020, during which women’s participation in the labour market increased significantly. Therefore, the actual difference in working hours between genders may be even greater for individuals with the same characteristics.

The descriptive statistics in this study of a specific cohort in 2008 show that higher education is positively correlated with wages and a decreasing trend in working hours, potentially due to possessing skills or abilities that increase productivity. This cohort represents middle-aged workers who have already invested significant time and resources into their careers, so these

summary statistics must be interpreted with that in mind. The findings reveal significant disparities in labour market outcomes between individuals with different education levels and genders, with higher-educated individuals earning more and working fewer hours. The gender gap in working hours narrows with increasing education. (References: Table 4, Table 2, Table 3.)

4 Empirical results

In the empirical results section, we analyse the relationship between human capital evolution and education and gender characteristics in the labour market. Our analysis aims to answer key questions, such as the impact of secondary education on wage age profiles and potential gender differences. We also explore how investing in human capital affects hour-age profiles and identify potential disparities and trends in the gender gap.

4.1 The estimation of wage age profile

According to the human capital theory, the wage-age profile reflects the relationship between an individual’s age and their wages, which is influenced by their investment in human capital over time. To explore this relationship, we use regression analysis with specification (1) and present the results in Table 5. The key parameters include β_{age^2} , which indicates the decline in wages over time and reflects depreciation, and β_{age} , which represents the return on investment in human capital over time. All the coefficients are statistically significant, providing valuable insights into the dynamics of the labour market and the role of human capital investments in shaping the wage-age profile.

If we focus on individuals with lower education (column (2) of Table 5), the results indicate that an increase of one year in age leads to an increase of $\approx 11\%$ in wages. On the other hand, for individuals with higher education (column (1)), an increase of one year in age leads to an increase of $\approx 12.6\%$ in wages. These findings are consistent with the human capital theory, which suggests that having a secondary level education instead of a lower secondary education leads to higher returns, while the depreciation rate remains the same for both groups. The low overall R^2 ($\approx 11\%$) is expected: with cohort and year fixed effects absorbing much of the variation, identification of the age–wage relationship comes from within-cohort, within-year variation rather

than from overall predictive fit. The R^2 therefore understates how well the specification captures the age profile of wages.

To visually inspect the results we show the predicted wages by cohorts.

In this section, we present a figure (see Figure 1) that visually depicts the relationship between education levels and wages across different age cohorts. The results are intriguing, as they suggest that there is a notable difference in the wage gap between older and younger cohorts. The figure shows that the wage gap is larger for the older cohort, while the younger cohorts have a narrower difference. This finding is consistent with the notion that higher educated individuals tend to spend part of their time endowment continuing to invest in their human capital during the early stages of their careers, while the lower educated tend to have more experience as they started working early, thus accumulating specialised skills. It is worth noting that this observation may indicate that specialised skills have a higher depreciation rate and a lower return in the long term. It would be interesting to extend the model to include two types of human capital assets: one with a higher short-term return and higher depreciation, and another that is more costly but has lower depreciation and then let the agent choose the optimal level of the two assets following the idea of Krueger and Kumar (2004).

The findings of our study support the theory that higher levels of human capital lead to higher wages. Specifically, Figure 1 demonstrates that individuals with higher levels of education consistently earn higher wages compared to those with lower education levels across all age cohorts. This observation highlights the importance of investing in human capital, particularly at the early stages of one's career, to enhance earning potential in the long term.

Furthermore, Figure 1 also shows that the wage gap between education levels widens over time. This suggests that younger cohorts have a smaller wage gap because the lower-educated individuals are more experienced in the labour market, while the higher-educated individuals are just starting. However, as time goes by, the market starts to remunerate better those individuals who have invested in higher human capital.

Building upon these findings, we extend our analysis to include the effects of gender on wages by conducting a regression analysis similar to the one before but with gender as a conditioning variable. This will allow us to examine the gender wage gap across different education levels and age cohorts, providing important insights into the dynamics of the labour market and the role of gender in wage determination.

The results presented in Table 6 indicate that there is clear evidence

of a lower return on education for women, with β_{age} for women (columns (2) and (4)) being lower than β_{age} for men (columns (1) and (3)). This effect is more pronounced for individuals with higher levels of education, as the marginal gap concerning age is wider, with $\Delta_{secondary} \approx 2.2\%$ per year for those with secondary education, and $\Delta_{lower\ secondary} \approx 1.4\%$ per year for those with lower secondary education. This result is consistent with previous research (Erosa et al., 2012) and implies a lower return on investment in human capital for women. Interestingly, there is no statistically significant difference in the depreciation of human capital over time between men and women: $\beta_{age^2} | male \approx \beta_{age^2} | female$, at approximately 0.83%.

The gender wage gap can be decomposed into two factors that widen it over time: the increase in wages as a result of acquiring working experience, represented by β_{age} , and the decrease in wages, represented by β_{age^2} .

These findings emphasise the importance of considering gender as a conditioning variable in wage regression analysis, as it provides valuable insights into the dynamics of the labour market and the role of human capital investments in shaping the wage-age profile. Moreover, the results suggest that women have lower remuneration for their human capital, which partially explains the wide difference in wages shown in Table 2. To better understand the difference in the evolution of wage-age profiles for women and men, we plot the predicted $\log(wage)$ comparing individuals with the same level of education but of different genders.

Figure 2 illustrates the male and female wage-age profiles conditional on education level across all cohorts, while Figure 3 restricts attention to cohort 6. We now turn to the same results for secondary and lower secondary school.

Figure 3 shows a larger gender gap for the low-educated that does not fade out as time passes. It is important to remember that the prediction of wages is based only on observables, so women who drop out of their job and stay at home are not taken into account; the gap that we observe therefore applies only to the women who stay in the market. Moreover, the wage gap is less accentuated for women with higher education, but this can be justified by the fact that the women who choose to remain in the market are those without childbearing obligations or those who are paid more.³ Furthermore,

³Studying women’s decision to drop out would help identify the reservation wage and thus instrument for the observability of wages. Consistent with this, a naive regression suggests that having a child raises female wages by about 8%; this is a selection artefact rather than a causal effect, since participation collapses after the first child, leaving in the sample only the positively selected mothers who earn more, alongside women without

the mean age in cohort 6 is 48, so Figure 3 approximates the gender gap only in the neighbourhood of this age. Regardless of the strong bias due to endogeneity of the extensive margin decision, the wage-age profiles for women remain lower than those for men, meaning that women have a lower return on a year of education than men.

Overall, the estimated returns to education are consistent with the human capital theory. The comparison here is between individuals with a lower secondary and a secondary education level; a comparison with individuals holding a bachelor’s degree would have been preferable, but it is not feasible because there are only ≈ 600 such observations, too few to obtain a significant estimate of the key parameters. Regarding gender differences, there is clearly a lower return on human capital for women compared to men, although this gap is strongly underestimated for the reasons set out above.

4.2 The estimation of hour age profile

Working hours are a fundamental variable in the human capital model, as they are the outcome of an optimisation process whereby individuals choose how much of their time endowment to allocate towards increasing their human capital, and how much to devote to working. In this sense, working hours can be seen as a proxy for investment in human capital, as individuals who invest more time in developing their skills and knowledge may have fewer hours available for paid work.

First, we will look at the difference in the hours-age profiles for individuals with different levels of education. Second, we will look at the difference in the hours-age profile as before but conditioning on gender. In Table 7, we estimate by OLS the third specification (3) for lower secondary and secondary education levels. Column (1) reports the regression for the group that has achieved a secondary school education level, while column (2) reports the regression for individuals who achieved only lower secondary school.

Our analysis shows that individuals with lower levels of education tend to start working earlier in life compared to those with higher education. Indeed, β_{age} is positive for those who spend more time in school and negative for those with lower education,⁴ meaning that for the higher educated the time endowment spent working increases over time at a rate of 3.1% per

children.

⁴Both are statistically significant at the 1% level.

year. The observation that the time spent working decreases over time at a rate of -0.7% per year for lower-educated individuals is in line with the idea that higher-educated individuals tend to increase their working hours over time as they invest in their human capital early in their careers, and gradually reduce their working hours as they gain experience. This pattern is consistent with the human capital model, which predicts that individuals with lower levels of education tend to start working earlier in life to maximise their immediate earnings, while those with higher education may delay their entry into the labour market as they invest in their human capital.

To visually inspect these results, we plot the predicted values for working hours by education level, separating each cohort (Figure 4). This allows us to see the trends in working hours across different levels of education for each cohort, providing further insight into the relationship between human capital and working hours.

In Figure 4, the red line represents the hours-age profiles for individuals with lower secondary school education. These individuals tend to start spending their time endowment working earlier compared to those with secondary education. Additionally, individuals with lower education have a higher hours-age profile, which decreases over time. The peak of the red group’s hours-age profile is reached in the early years of their working career, and then the curve slightly decreases over time. In contrast, for those with higher education, the peak is reached between 40 and 50 years old, and then starts to decrease. It is important to emphasise that individuals with higher education spend less time working for all years, and this trend is observed in all cohorts.

To examine gender differences in the hours-age profile, we estimated the same regression model as before (specification (4)), but stratified by gender. Table 8 displays the results, which indicate that the slope of the hours-age profile is negative for individuals with higher education levels, as in the results without conditioning on gender. However, the negative slope is flatter for women across all education levels, with a difference of only $\Delta_{female-male}\beta_{age} \approx 0.1\%$ per year, suggesting that women tend to increase their working hours more slowly than men. Nevertheless, low-educated women have steeper negative hours-age profiles ($\beta_{age} | female \approx -1.7\%$ per year) and a wider gender gap ($\Delta_{female-male}\beta_{age} \approx 1.9\%$ per year) than highly educated women, implying that low-educated women tend to decrease their labour market participation faster than their highly educated counterparts.

We visually compared the predicted working hours by cohort for men and

women with different levels of education. Figure 5 displays the predicted values of specification (4), where men with higher education are in orange, men with lower education are in red, women with lower education are in blue, and women with higher education are in green. We also plot the predicted values for the middle-aged cohort 7 in Figure 6.

The figure illustrates that men have higher hours-age profiles than women across all cohorts, indicating that women have lower participation in the intensive margin compared to men, regardless of educational attainment. Additionally, for women with lower education, the hours-age profile declines rapidly in the initial stages of their working lives, particularly between ages 20 and 30. Consequently, the gap between low-educated females and males widens considerably during this period, reaching its peak at around age 40. Conversely, for women who stay in school longer, there is an increase in participation during the early stages of their careers, between ages 20 and 30, when the hours-age profile reaches its maximum. Subsequently, the participation rate either declines or remains stable. However, it is worth noting that the decrease in hours worked for women is significantly underestimated, since the probability of dropping out of the labour force after having the first child is around 30% for individuals with lower and secondary school education, as demonstrated in Bratti et al. (2005), which implies that a large proportion of female labour force participation is unobservable.

Moreover, the gender gap in the hourly-age profile reaches its peak at around age 50 for those with higher education, which is consistent with the idea that women who stay in school longer tend to have their first child later, delaying their childbearing years until later in their working lives. The gap begins to decrease slightly after age 50 but remains significant until the end of their working careers. Interestingly, the gender gap in the hourly-age profile is wider for those who spent more time in schooling, as observed in the younger cohort. One possible explanation is that women who pursue higher levels of education may face additional pressure to conform to traditional gender roles, such as being a caregiver for children and elderly family members, which can lead to a reduction in their labour force participation or hours worked. This can be particularly true during the early stages of their careers when they are starting families or establishing themselves in their chosen fields. On the other hand, men who pursue higher education may face fewer societal expectations to prioritise caregiving responsibilities over their careers. Thus, they may be able to invest more time and effort into their careers.

The hours-age profiles for lower-educated women were consistently lower

than for men across all cohorts, indicating that this gender gap persists over time. However, for higher-educated individuals, the gender differences were not apparent (as shown in Figure 5).

Overall, our empirical findings are in line with the human capital theory, as we found a positive and statistically significant effect of education on human capital remuneration and a concave wage-age profile as predicted by the model with depreciation $\Delta > 0$. Additionally, the hours-age profile supports the human capital model's predictions, with higher-educated individuals having an increasing hours-age profile, and lower-educated individuals having a decreasing one. Furthermore, our analysis of the gender gap in the hours-age profile shows that women tend to have lower participation in the intensive margin compared to men, regardless of their level of educational attainment. The gap is wider for lower-educated women, and it decreases slightly for higher-educated women after the age of 50, which is consistent with the fact that they tend to have their first child later and shift childbearing years later in their working life. It is also interesting to note that the gender gap in the hourly-age profile at the beginning of the career is wider for those who spent more time in schooling, as seen in the younger cohort. These findings provide insights into the complex interplay between education, gender, and labour market outcomes.

5 Conclusions

In conclusion, this paper has provided an analysis of the gender wage gap in a European country, focusing on the role of education and human capital. The empirical findings are consistent with the human capital theory, as we observe a positive statistically significant effect of education on the remuneration of human capital and a concave shape of the wage-age profile, in line with the model with $\Delta > 0$.

Furthermore, the analysis shows that the gender wage gap widens over time, mainly due to differences in the remuneration of human capital between men and women. Women have a lower remuneration of human capital, even for the same level of education, which can be partly explained by differences in working hours, as women tend to invest less time in developing their skills and knowledge.

The results also indicate that individuals with lower levels of education start working earlier, to maximise their immediate earnings, while those with

higher levels of education may take longer to start working as they invest in their human capital (Ben-Porath, 1967). This is reflected in the hours-age profile, which decreases over time for those with lower education, while it increases for those with higher education.

However, it is important to note that the analysis has certain limitations. Firstly, the restriction imposed on cohort fixed effects and year fixed effects may not capture all the heterogeneity in the data. Secondly, the analysis only focuses on a single country, and the results may not be generalizable to other contexts. Finally, the study does not account for the potential impact of discrimination and other factors that may contribute to the gender wage gap.

Overall, this study provides insights into the factors that contribute to the gender wage gap and highlights the importance of education and human capital in determining individuals' earnings. Further research is needed to explore the role of other factors, such as discrimination and family responsibilities, and to investigate the effectiveness of policies aimed at reducing the gender wage gap.

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6 Tables and figures

Table 1: Freq. density of cohorts			
Cohort	N.obs	Freq.Perc.	Cum.
1	42	0.04	0.04
2	619	0.59	0.63
3	1816	1.74	2.37
4	3333	3.20	5.57
5	6869	6.58	12.15
6	9495	9.10	21.26
7	11401	10.93	32.19
8	11971	11.48	43.66
9	13347	12.79	56.46
10	14240	13.65	70.11
11	10862	10.41	80.52
12	8222	7.88	88.40
13	6291	6.03	94.43
14	4291	4.11	98.55
15	1517	1.45	100.00
Total	104316	100	

Table 2: Descriptive statistics wages

Education level	Male	Female	Diff.
Lower Secondary	51.71 (33.06)	48.06 (40.60)	3.642061*** (.9713213)
Secondary	124.03 (91.37)	113.09 (73.14)	5.360341*** (.5809988)
Diff.	-18.47816*** (.7922257)	-16.75988*** (1.07596)	

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 3: Descriptive statistics annual working hours

Education level	Male	Female	Diff
Lower Secondary	1,952.12 (393.85)	1,642.11 (537.70)	310.0051*** (12.13044)
Secondary	1,811.48 (484.11)	1,507.51 (491.03)	253.783*** (4.887414)
Diff.	27.01869 *** (6.180364)	-29.20347*** (11.81991)	

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 4: Descriptive statistics of cohort=10 and year=2008

	no edu.	primary	lower second.	second. 3y	second.
Mean of log (wage)	3.61 (.)	3.63 (0.06)	3.67 (0.03)	3.69 (0.02)	3.75 (0.02)
Mean of work hour	1,920.00 (.)	1,886.70 (116.59)	1,811.86 (50.03)	1,780.06 (31.46)	1,678.31 (38.77)
mean_y	71261.03 (.)	71261.03 (0.00)	71261.03 (0.00)	71261.03 (0.00)	71261.03 (0.00)
Female	0.00 (.)	0.26 (0.45)	0.40 (0.49)	0.55 (0.50)	0.66 (0.48)

Table 5: Regression on $\log(wage)$

	(1)Sec.	(2)Lower sec.
Age	0.126*** (7.07)	0.109*** (17.45)
Age ²	-0.000871*** (-4.70)	-0.000832*** (-12.96)
Constant	-0.199 (-0.25)	0.0380 (0.08)
Year fixed effect	Yes	Yes
Cohort fixed effect	Yes	Yes
Observations	10088	28968
R^2 overall	0.0595	0.1072

t statistics in parentheses

Regression on $\log(wage)$ for the secondary school group (1) and the lower secondary group (2)

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 6: Regression on $E[\log(wage) \mid female, educ]$

	(1)M., H. educ.	(2)F. H. educ.	(3)M., L. educ.	(4)F., L. educ.
Age	0.13710*** (0.03092)	0.11522*** (0.02028)	0.11396*** (0.00794)	0.10093*** (0.01015)
Age ²	-0.00090*** (0.00032)	-0.00083*** (0.00021)	-0.00089*** (0.00008)	-0.00070*** (0.00010)
Cohort f.e.	Yes	Yes	Yes	Yes
Years f.e.	Yes	Yes	Yes	Yes
Constant	-0.83300 (1.22242)	0.44851 (1.11800)	-0.12925 (0.54144)	0.69487 (1.17340)
Observations	4593	5495	19045	9923
R^2 overall	0.0499	0.0777	0.1085	0.1114

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 7: Regression $E[\log(hours) \mid educ]$

	(1) Sec.	(2) Lower sec.
Age	0.031*** (0.003)	-0.007*** (0.000)
Age ²	-0.000*** (0.000)	0.000*** (0.000)
Cohort f.e.	Yes	Yes
Years f.e.	Yes	Yes
Constant	6.786*** (0.166)	7.833*** (0.020)
Observations	10088	28968
R^2 overall	0.0345	0.3897

Column (1) is the secondary school group (educ= 5), column (2) the lower secondary group (educ= 3).

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 8: Regression table $E[\log(hours) \mid female, educ]$

	(1)[1, 0]	(2)[1, 1]	(3)[0, 0]	(4)[0, 1]
Age	0.00788 (0.00567)	0.00706 (0.00597)	0.00202 (0.00163)	-0.01676*** (0.00357)
Age ²	-0.00009 (0.00006)	-0.00008 (0.00006)	-0.00006*** (0.00002)	0.00014*** (0.00004)
Cohort f.e.	Yes	Yes	Yes	Yes
Years f.e.	Yes	Yes	Yes	Yes
Constant	7.43107*** (0.22349)	6.78971*** (0.33563)	7.69220*** (0.11123)	7.47544*** (0.41583)
Observations	4593	5495	19045	9923
R^2 overall	—	—	—	—

The regressions are conditional on education and gender: each column is indexed by $[educ, female]$, where $educ=1$ denotes a secondary school education level and $educ=0$ a lower secondary one.

R^2 overall is not reported: the values carried in an earlier draft were erroneously duplicated from Table 6 and these regressions await re-estimation.

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

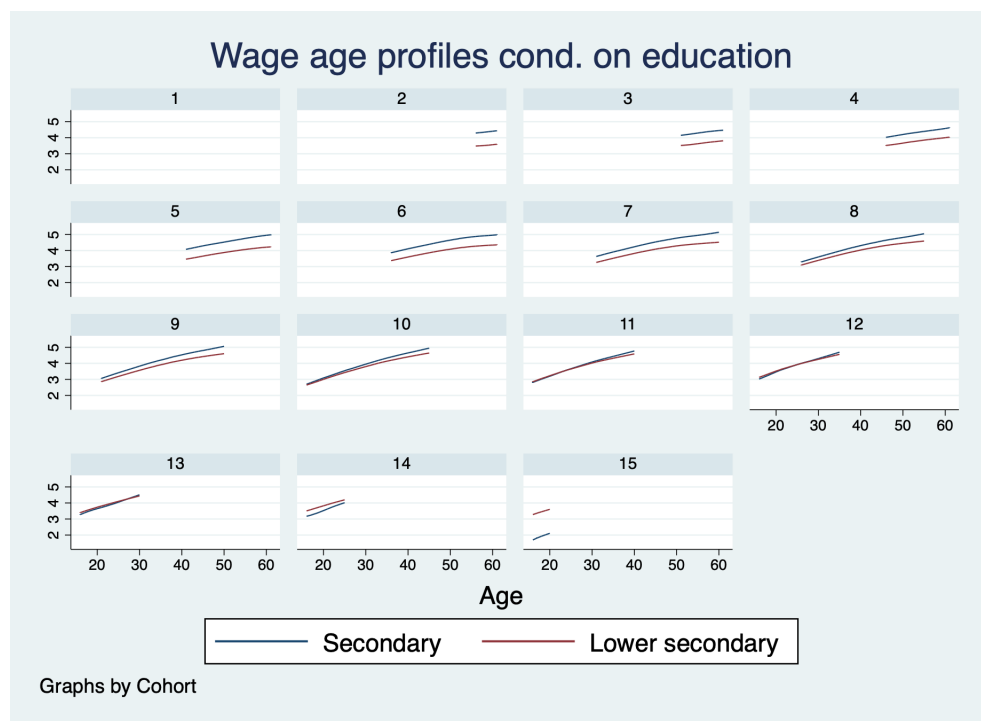


Figure 1: Predicted wage-age profiles by cohort

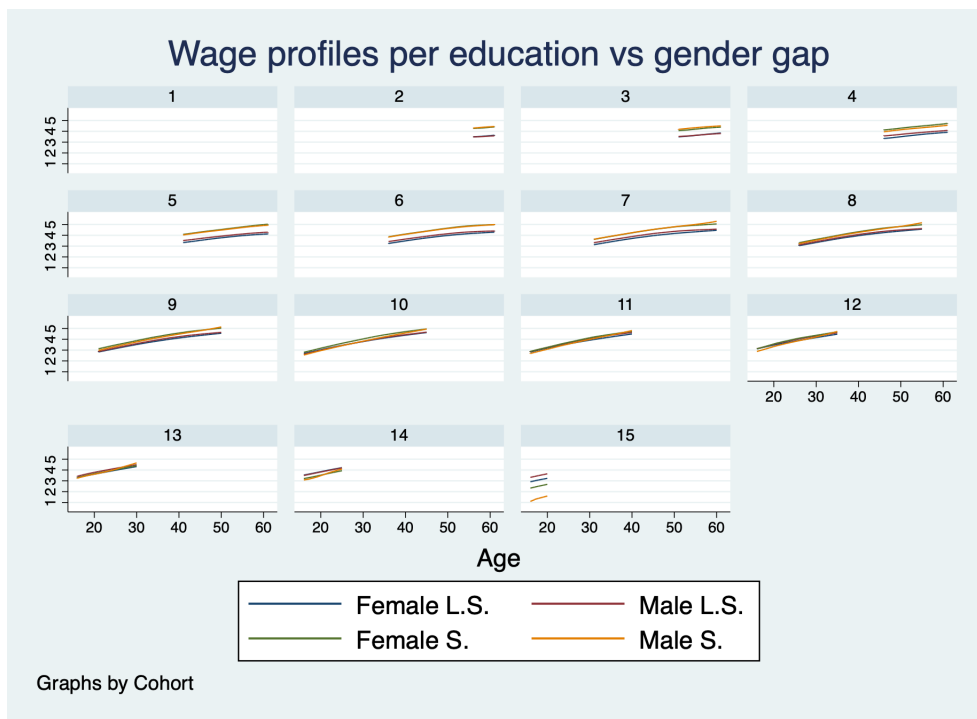


Figure 2: Gender gap in the wage-age profiles by education level

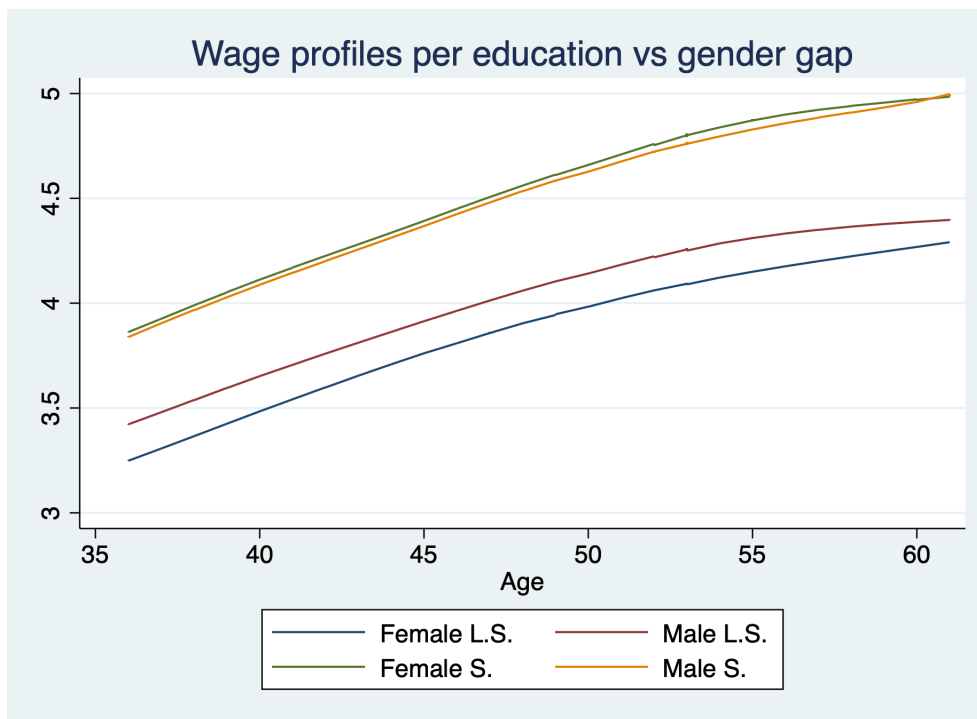


Figure 3: Gender gap in the wage-age profiles by education level, cohort 6

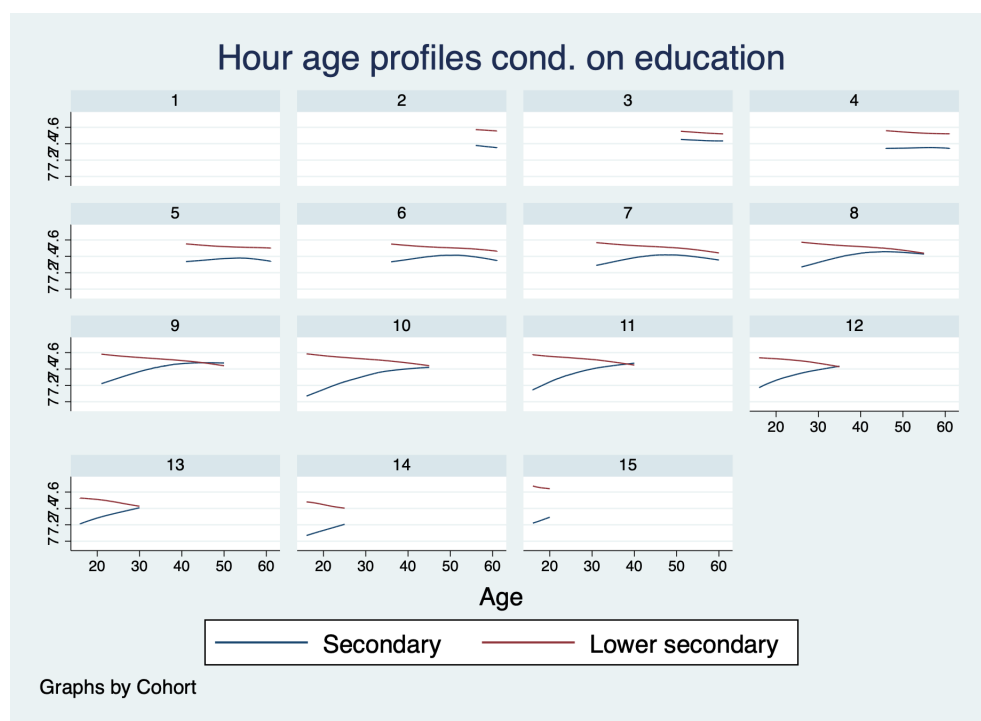


Figure 4: Hours-age profile

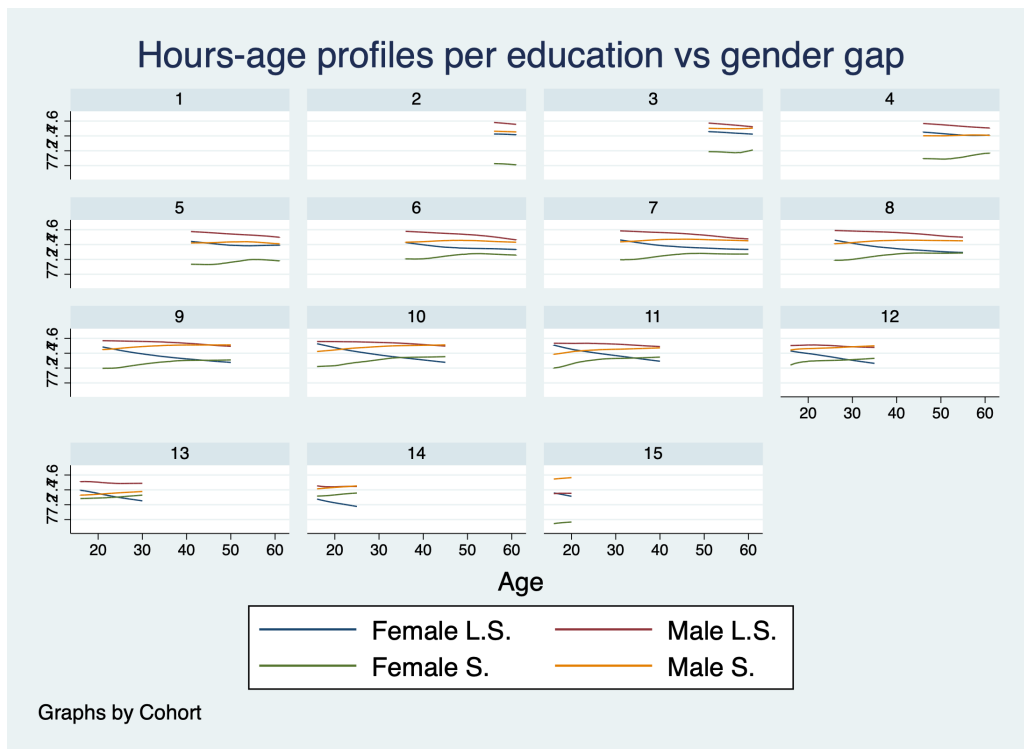


Figure 5: Hours-age profile and gender gap

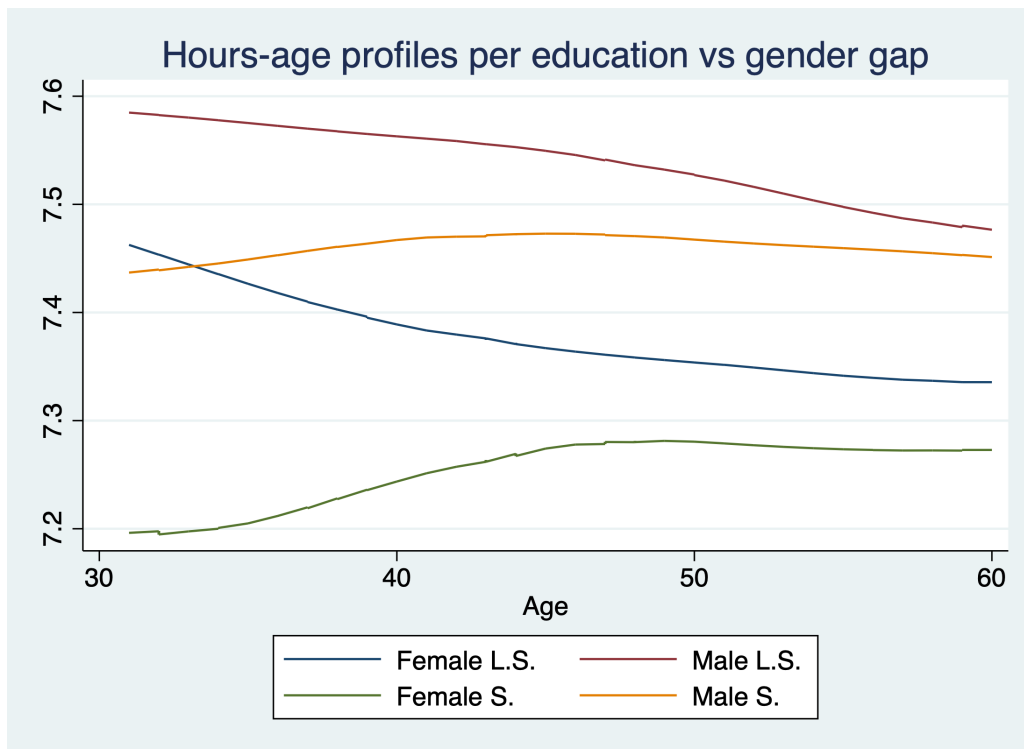


Figure 6: Hours-age profile and gender gap, cohort 7